

# TOWARDS REVERSIBLE MODEL MERGING FOR LOW-RANK WEIGHTS

Anonymous authors

Paper under double-blind review

## ABSTRACT

Model merging aims to combine multiple fine-tuned models into a single set of weights that performs well across all source tasks. Prior work has shown that merging can approximate the performance of individual fine-tuned models, but it largely overlooks the increasingly common regime in which models are compressed into low-rank representations—either through low-rank adaptation (LoRA) or post-training singular value decomposition (SVD). This paper makes three contributions to model merging in this low-rank regime. **First**, we identify and characterize a failure mode: applying conventional merging methods to low-rank weights leads to severe, often catastrophic, performance degradation in the merged model. **Second**, motivated by this phenomenon, we propose a fundamentally different formulation of merging. Instead of collapsing all adapters into one set of weights, we construct a compact basis (e.g., the equivalent of holding two or more models) from which the original task-specific models can be recovered via linear combination. This reframes merging as generating a *reconstruction-capable model space* rather than a single merged model, and crucially allows us to “revert” to each individual model when needed—recognizing that no merged model can consistently outperform one specialized for its task. **Third**, building on this insight, we introduce Reversible Model Merging (RMM), an efficient, data-free, and flexible method that gives a closed-form solution for selecting both the optimal basis of model weights and the task-specific coefficients used for linear combination. Extensive experiments across diverse datasets and model scales confirm each contribution: RMM consistently outperforms existing merging approaches and preserves the performance of low-rank compressed models by a significant margin, while exposing a tunable storage–accuracy trade-off.

## 1 INTRODUCTION

Parameter-efficient fine-tuning methods, such as Low-Rank Adapters (LoRA) (Hu et al., 2021), have become a cornerstone for adapting large language models to diverse downstream tasks without incurring the full cost of retraining. In parallel, model compression techniques such as post-training singular value decomposition (SVD) truncation have been widely adopted to reduce storage and inference costs, offering an additional dimension of efficiency (Ryu et al., 2023; Ping et al., 2024; Wang et al., 2025b;a). Alongside these advances, model merging has emerged as a promising paradigm for consolidating the knowledge of multiple task-specific models into a single unified model, and recent methods—ranging from simple parameter interpolation to optimization-based techniques—have shown that merged models can perform competitively across source tasks without access to the original training data (Matena & Raffel, 2022; Yang et al., 2024; Yadav et al., 2023; Wei et al., 2025). This paper studies model merging at the intersection of these two trends—merging models that are *already* compressed into low-rank form—a setting that, as we show, breaks the assumptions underlying existing merging methods.

Despite this progress, an important limitation remains largely overlooked, and it motivates our first contribution. Even the most sophisticated merging strategies consistently fail to match the performance of the original individual fine-tuned models, and this performance gap becomes especially pronounced—often catastrophic—when the models being merged have already been compressed into low-rank form, whether via LoRA fine-tuning or post-training SVD truncation. The root cause is structural: while merging lets a single model handle multiple tasks, it collapses distinct task-

specific representations into one parameter space, inducing interference between tasks and eroding the specialized adaptations learned by each model (Gargiulo et al., 2025). Under low-rank compression, where representational capacity is already scarce, this collapse is far less forgiving.

Why does this setting matter? Although low-rank adapters and SVD-truncated models are individually compact, their proliferation introduces a new scalability bottleneck. In practical multi-task, federated, or continual learning scenarios, it is common to accumulate dozens or even hundreds of task-specific low-rank adapters, each requiring separate storage and management coordination. Retaining all of them erodes the very storage savings that low-rank parameterization was designed to provide and makes deployment cumbersome. The problem we target, therefore, is the design of merging strategies that operate directly on low-rank representations—not only to consolidate storage and reduce redundancy, but also to enable more efficient sharing, deployment, and transfer of compressed models across tasks.

**Contribution 1: diagnosing the failure of merging under low-rank compression.** To ground the problem, we first investigate how existing merging techniques behave when applied to low-rank compressed models. We focus on *deltas*, i.e., the weight differences between each fine-tuned model and the shared pre-trained initialization. Conventional merging strategies attempt to combine these deltas into a single set of parameters under an assumption of linear compatibility. We show empirically that merging low-rank deltas directly leads to catastrophic performance degradation, and we trace this failure to two concrete factors: (i) the limited expressive capacity inherent in low-rank representations, and (ii) the misalignment of task-specific subspaces, each optimized independently, which produces severe interference when combined. This diagnosis is what motivates the reformulation below.

**Contribution 2: reframing merging as constructing a reconstruction-capable model space.** Motivated by this diagnosis, we take a step back and ask: *What if the goal of merging were not to produce a single final model? What if we relaxed the problem to allow retaining more than one model? What if we preserved the ability to reconstruct the original models on demand?* Instead of collapsing multiple low-rank adapters into a single set of weights, we propose to maintain a compact set of weights that serves as a shared basis from which each original model can be recovered through linear combination. This reframing opens an underexplored design space: rather than compressing all information into one model, we preserve complementary representational directions across a small set of basis models, enabling post-merging “reversion” to the original adapters. In short, we challenge the prevailing assumption that merging must terminate in a single model, and instead redefine merging as the construction of a reconstruction-capable model space.

**Contribution 3: Reversible Model Merging (RMM), a closed-form method.** Building on this reconstruction-driven perspective, we develop a theoretical framework that characterizes the optimal basis for minimizing reconstruction error, and turn it into a concrete algorithm: Reversible Model Merging (RMM). We formalize merging as a basis-selection problem whose objective is to identify a compact set of task vectors from which all original models can be accurately reconstructed. Within this framework, RMM admits a closed-form solution that simultaneously (i) identifies an optimal set of basis task vectors and (ii) determines the task-specific coefficients for the linear combination that recovers each individual task adaptation. RMM is data-free and requires no retraining. Empirically, we validate it across a broad range of datasets and model configurations, and it consistently surpasses conventional merging strategies by a significant margin while preserving the efficiency benefits of low-rank parameterization—underscoring its practicality for large-scale multi-task adaptation. It further exposes a flexible storage–performance trade-off through a single tunable hyperparameter that controls the number of models retained. In summary, the three contributions above—the failure diagnosis, the reconstruction-capable reformulation, and the closed-form RMM method—together reframe how compressed models can be merged, stored, and recovered.

## 2 RELATED WORK

Model merging has emerged as a powerful strategy for consolidating knowledge from multiple fine-tuned models. Existing methods broadly fall into two categories: parameter-space manipulations that require no data, and optimization-based approaches that leverage data or auxiliary objectives. We review representative techniques from both lines below, emphasizing that—regardless

of category—they all share the goal of producing a *single* merged model, the very assumption our reconstruction-capable reformulation departs from.

**Task Arithmetic:** The foundational idea in model merging is task arithmetic (Ilharco et al., 2023), which defines a delta as the difference between a fine-tuned model’s parameters and its pre-trained initialization. Its key insight is that such deltas behave like modular building blocks: they can be added to a base model to impart new abilities, subtracted to suppress unwanted behaviors, or scaled and combined to blend multiple capabilities. This intuitive linear framework established the basis for composing model behaviors directly in weight space, and it is the starting point our method also builds on.

**DARE:** Simple delta addition can be effective, but it often fails when merged models contain conflicting or overlapping parameter updates. To address this, Yu et al. (2024) introduced DARE (Drop and Rescale), built on aggressive randomized pruning. DARE discards the vast majority of parameter changes—up to 90–99%—on the premise that most updates are inessential and may even interfere with knowledge from other models, and then amplifies the remaining updates to preserve the overall scale of the modification. The net effect is to retain core task-specific modifications while minimizing interference.

**TIES:** In contrast to DARE’s randomized pruning, TIES-merging offers a deterministic framework for mitigating parameter conflicts (Yadav et al., 2023). It proceeds in three phases—Trim, Elect Sign, and Merge—each targeting interference across models: it first retains only the largest-magnitude parameters so that the most influential fine-tuning updates survive; it then, at each parameter position, evaluates the sign of updates across models and enforces agreement by discarding contributions that oppose the majority sign; and it finally averages the sign-consistent parameters into the merged representation. This consensus-driven approach reduces destructive interference and yields a more stable merged model. Sharing the same interference-reduction motivation, EMR-Merging (Elect, Mask, and Rescale) elects a unified model and attaches task-specific binary masks and rescalers to shift that model toward each individual task (Huang et al., 2024). We note that all of these methods still commit to a single merged model—the limitation our approach removes.

**Other Merging Methods:** Beyond purely parameter-space manipulations, several approaches rely on optimization procedures or access to a small dataset. Model soups (Wortsman et al., 2022) refine task-arithmetic merging weights by minimizing loss on validation data, steering the interpolation toward better generalization, while Fisher-weighted averaging (Matena & Raffel, 2022) incorporates second-order information to weight parameters by their estimated importance for task performance. Other lines of work cast merging explicitly as an optimization problem: RegMean (Jin et al., 2023) and AdaMerging (Yang et al., 2024) adaptively search for task-dependent merging coefficients by solving small-scale objectives—on the discrepancy between merged- and individual-model predictions, and on the entropy of predictions for unlabeled data, respectively—and a recent method minimizes the loss difference between merged and individual models via a first-order Taylor expansion that yields a tractable surrogate objective (Wei et al., 2025). These approaches can improve task-specific performance, but they require labeled or unlabeled data, additional optimization, or both—limiting their use where merging must be done without retraining or data. Against this backdrop, our contribution is distinctive on two axes: RMM is fully data-free and closed-form, *and* it abandons the single-model target that all of the above share.

### 3 PRELIMINARIES

We begin by formalizing the conventional merging problem that our contributions later depart from. Consider consolidating a collection of fine-tuned models into a single unified model via model merging. Let  $\{\theta_1, \dots, \theta_n\}$  denote  $n$  models fine-tuned on  $n$  distinct tasks. A merging algorithm  $\mathcal{M}$  combines these models into a single set of weights  $\theta_{\mathcal{M}} = \mathcal{M}(\{\theta_i\}_{i=1}^n)$ . The objective is to identify a merging procedure  $\mathcal{M}$  that maximizes the average validation performance across all tasks:

$$\max_{\mathcal{M}} \frac{1}{n} \sum_{i=1}^n \text{Perf}(\theta_{\mathcal{M}}, \mathcal{D}_i), \quad (1)$$

where  $\mathcal{D}_i$  denotes a held-out validation set for task  $i$ , and  $\text{Perf}$  denotes any suitable performance metric (e.g., accuracy or F1 score). It is standard to assume that all fine-tuned models originate from

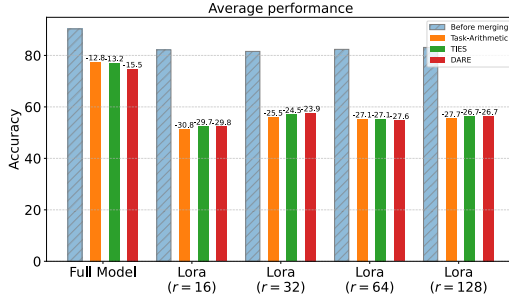


Figure 1: Evidence for Contribution 1 (LoRA): average task performance of individual models (before merging) versus merged models across merging methods. The widening gap once models are LoRA-compressed—shown for full fine-tuning and ranks  $r \in \{16, 32, 64, 128\}$ —is the failure mode that motivates RMM.

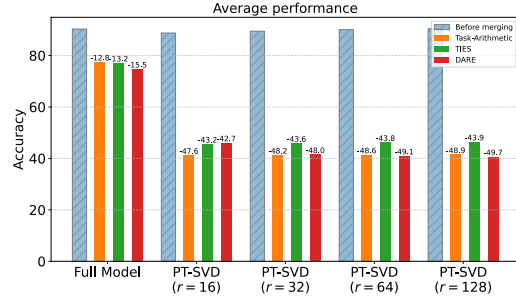


Figure 2: Evidence for Contribution 1 (PT-SVD): average task performance of individual models (before merging) versus merged models across merging methods. As with LoRA, PT-SVD compression—shown for full fine-tuning and ranks  $r \in \{16, 32, 64, 128\}$ —sharply widens the gap between merged and individual models.

the same pretrained initialization  $\theta_{\text{pre}}$ . For any layer  $l$ , the update (or delta) of model  $i$  relative to the base model is defined as  $\Delta_i^l = \theta_i^l - \theta_{\text{pre}}^l$ , where  $\Delta_i^l \in \mathbb{R}^{m \times d}$  denotes the weight difference for a layer of dimensions  $m \times d$ .

### 3.1 LOW-RANK COMPRESSED MODELS

Because our contributions concern merging *compressed* models, we now make the low-rank setting precise. In practice, it is common to compress fine-tuned models into low-rank form to reduce storage, training, and inference costs. Two widely adopted compression strategies are:

**LoRA fine-tuning:** During training, each layer is augmented with trainable low-rank matrices  $A \in \mathbb{R}^{m \times r}$  and  $B \in \mathbb{R}^{r \times d}$ , where  $r \ll \min(m, d)$ . The pretrained weights remain frozen, and only  $A$  and  $B$  are optimized.

**Post-training SVD truncation (PT-SVD):** After standard fine-tuning, the update  $\Delta_i^l$  is decomposed via SVD and truncated to retain only the top- $r$  singular components, yielding a low-rank approximation  $\hat{\Delta}_i^l = A_i^l B_i^l$ , for  $A_i^l \in \mathbb{R}^{m \times r}$  and  $B_i^l \in \mathbb{R}^{r \times d}$  where  $r \ll \min(m, d)$ .

In both cases, the fine-tuned update for layer  $l$  in model  $i$  is represented in low-rank form  $\hat{\Delta}_i^l = A_i^l B_i^l$ , where  $A_i^l \in \mathbb{R}^{m \times r}$  and  $B_i^l \in \mathbb{R}^{r \times d}$ , with  $r$  denoting the rank of compression. Thus, the merging problem becomes one of combining low-rank updates across tasks. The goal is to construct a merged model that integrates the compressed deltas  $\hat{\Delta}_i = \{\hat{\Delta}_i^l\}_{l=1}^L$  across all the tasks:

$$\hat{\theta}_{\mathcal{M}} = \theta_{\text{pre}} + \mathcal{M}(\{\hat{\Delta}_i\}_{i=1}^n), \quad \max_{\mathcal{M}} \frac{1}{n} \sum_{i=1}^n \text{Perf}(\hat{\theta}_{\mathcal{M}}, \mathcal{D}_i), \quad (2)$$

where  $L$  denotes the total number of layers. This low-rank-merging problem is the setting for all three of our contributions. As a first step, the next section (delivering Contribution 1) investigates how conventional merging strategies behave in it—and shows that they break down.

## 4 MOTIVATING OBSERVATION

This section delivers our first contribution: a concrete diagnosis of why conventional merging fails on low-rank compressed models. Building on the setting of the previous section, we consider for each layer  $l$  the two natural extensions of merging examined in prior work (Tang et al., 2025; Zhao et al., 2025; Zheng et al., 2025; Zhang et al., 2025):

- **Combined merging:**

$$\hat{\theta}_{\mathcal{M}}^l = \theta_{\text{pre}}^l + \mathcal{M}(\{A_i^l B_i^l\}_{i=1}^n),$$

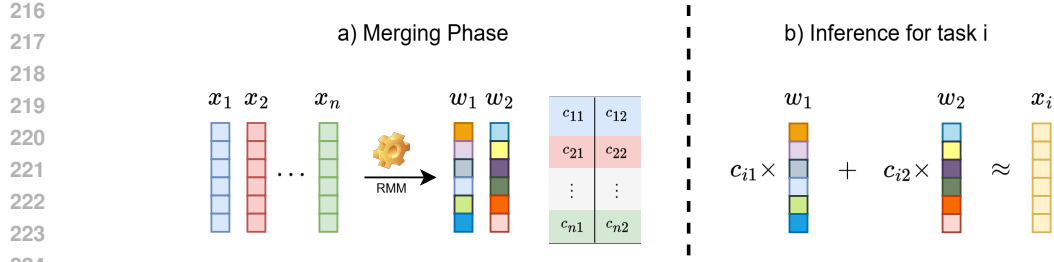


Figure 3: Overview of RMM, the paper’s third contribution. **Left (merging phase):** we compute a compact set of basis directions (two in this example) and their associated task-specific coefficients. **Right (inference phase):** a target model  $i$  is reconstructed—i.e., “reverted” to—by linearly combining the basis vectors with the corresponding coefficients. The two-phase design is what makes merging reversible.

where each low-rank update is first reconstructed into its full-rank form before merged.

- **Separate merging:**

$$\hat{\theta}_{\mathcal{M}}^l = \theta_{\text{pre}}^l + \mathcal{M}(\{\mathbf{A}_i^l\}_{i=1}^n) \mathcal{M}(\{\mathbf{B}_i^l\}_{i=1}^n),$$

where the low-rank factors are merged independently before recomposition.

Both approaches are theoretically valid, but the combined strategy is impractical: storing the merged update in combined form requires holding an  $m \times d$  matrix, whereas the separate approach stores only an amount equivalent to one of the low-rank factors  $\mathbf{A}_i^l \in \mathbb{R}^{m \times r}$  and  $\mathbf{B}_i^l \in \mathbb{R}^{r \times d}$ . Since  $r(m + d) \ll md$  for  $r \ll \min(m, d)$ , the combined approach’s memory overhead defeats the very purpose of compression. Concretely, for a layer with  $m = d = 1024$  and rank  $r = 16$ , the combined approach stores a full update of  $md = 1,048,576$  parameters while the separate approach stores only  $r(m + d) = 16 \times (1024 + 1024) = 32,768$ .

This is a **32× storage reduction**: the combined approach entirely undermines the efficiency benefits of low-rank compression. We therefore adopt the separate strategy as the only viable option for merging compressed models in practice, and it is the setting for the failure we report next.

To test this empirically, we applied several existing merging baselines to low-rank compressed models under varying compression ratios, using the separate approach above. With RoBERTa-base (Zhuang et al., 2021) as the pre-trained model, we ran experiments on 5 GLUE tasks (Wang et al., 2018) (details in Appendix A.3). The result, visible in Figures 1 and 2, is stark: merging compressed models causes a *catastrophic drop* in average task performance relative to merging full-rank models. Our diagnosis attributes this degradation to two factors: (i) **Amplified task interference**: compression discards much of the representational capacity, leaving low-rank models more susceptible to destructive interference when merged; (ii) **Loss of coupling between  $\mathbf{A}$  and  $\mathbf{B}$** : conventional merging treats the low-rank factors  $\mathbf{A}_i^l$  and  $\mathbf{B}_i^l$  as independent, ignoring the coupled nature of their product and producing structural misalignment in the merged updates. Together, these factors severely impair merged low-rank models and point to a deeper incompatibility between traditional merging and compressed representations—the gap that motivates our second and third contributions.

This diagnosis leads to the critical insight behind our second contribution: under conventional strategies, no merged model can reliably recover the task-specific performance of the original low-rank adapters. It raises a natural question: *Can we instead reconstruct the original compressed models—or approximate them closely—by retaining a compact set of basis models, perhaps equivalent to two or more adapters depending on the storage budget?* Rather than forcing all updates into a single representation, we explore an alternative design space in which merging constructs a *reconstruction-capable model space*. The next section formalizes this perspective and introduces our method, *Reversible Model Merging* (RMM).

## 5 REVERSIBLE MODEL MERGING

We now present our third contribution in full. The empirical failures diagnosed above expose a fundamental limitation: no single merged model can faithfully preserve the task-specific perfor-

mance of all low-rank compressed models. RMM responds by abandoning the goal of merging all models into one set of weights. Instead, it constructs a small set of shared components—basis models—from which every original task model can be approximately reconstructed on demand. This is the concrete realization of the reconstruction-capable model space introduced as Contribution 2.

Concretely, rather than forcing all  $n$  task-specific models into one merged solution  $\hat{\theta}_{\mathcal{M}}$ , RMM retains a compact set of  $p$  merged components, with  $2 \leq p \ll n$ . These span a low-dimensional subspace capturing the essential variations across tasks, enabling efficient per-model recovery. The single hyperparameter  $p$  is the knob behind our storage–accuracy trade-off: increasing it improves reconstruction fidelity at the cost of memory.

The rest of this section makes the construction precise. For each layer  $l$ , we focus on the  $r$ -dimensional vectors inside the low-rank delta matrices  $A_i^l$  and  $B_i^l$ —the rows of  $A_i^l$  and the columns of  $B_i^l$ , where  $r$  is the compression rank. We treat each such row or column as an independent *task vector* and call a specific row or column index a *task vector position*. For any fixed position, we collect the corresponding task vectors across all  $n$  models, denoting the one from model  $i$  by  $x_i \in \mathbb{R}^r$ . RMM is then applied independently at each position: we compress the set  $\{x_i\}_{i=1}^n$  by projecting it into a shared basis so that each vector can be reconstructed later. Per layer there are  $m + d$  positions in total— $m$  rows from  $A_i^l$  and  $d$  columns from  $B_i^l$ —and the procedure runs separately on each.

To this end, we seek a set of basis vectors  $\{w_1, \dots, w_p\}$  with each  $w_j \in \mathbb{R}^r$  for  $1 \leq j \leq p$ . Their span defines a subspace  $W = [w_1 \dots w_p] \in \mathbb{R}^{r \times p}$  into which each task vector  $x_i$  is projected, and we require the projection to reconstruct  $x_i$  as accurately as possible. We collect the reconstruction weights in a coefficient matrix  $C \in \mathbb{R}^{n \times p}$ , whose  $i$ -th row  $[c_{i1}, \dots, c_{ip}]$  gives the scalars that linearly combine the basis vectors to reconstruct  $x_i$ . The objective—which formalizes “reversibility”—is to choose  $W$  and  $C$  that minimize the total reconstruction error over all  $n$  task vectors:

$$\min_{W, C} \sum_{i=1}^n \left\| x_i - \sum_{j=1}^p c_{ij} w_j \right\|_2^2. \quad (3)$$

This formulation stores  $rp$  parameters for the basis and  $np$  for the coefficients, totaling  $p(r + n)$ —a significant reduction from the  $rn$  parameters needed to store all task vectors directly, and the source of RMM’s efficiency. As noted,  $p$  is the tunable hyperparameter controlling the performance–storage trade-off; Figure 3 illustrates the case  $p = 2$ . The optimization in equation 3 can be rewritten in matrix form as:

$$\min_{W, C} \|X - CW^\top\|_F^2, \quad (4)$$

where  $X = [x_1 \dots x_n]^\top \in \mathbb{R}^{n \times r}$  stacks all task vectors. For any  $W$ , the optimal  $C$  is the least-squares solution  $C^* = XW(W^\top W)^{-1}$ . Assuming orthonormal basis vectors ( $W^\top W = I_p, p < r$ ), the optimization reduces to:

$$\min_W \|X - XWW^\top\|_F^2 \quad \text{s.t.} \quad W^\top W = I_p. \quad (5)$$

In this formulation we store  $W \in \mathbb{R}^{r \times p}$  and  $XW \in \mathbb{R}^{n \times p}$  (here  $XW$  plays the role of  $C$  in equation 4). The key payoff of Contribution 3 is that this problem admits a closed-form optimum, stated next.

**Theorem 1** (Optimal Solution to equation 5). *Let  $X \in \mathbb{R}^{n \times r}$  be zero-centered and  $p < \min\{r, n\}$ . The optimal basis  $W^*$  that minimizes equation 5 is given by the top- $p$  eigenvectors of the sample covariance matrix  $X^\top X$ . Equivalently,  $W^*$  consists of the top- $p$  right singular vectors of  $X$ .*

The proof of Theorem 1 appears in Appendix A.1. This closed-form, data-free solution is precisely what distinguishes RMM from the optimization- and data-dependent baselines surveyed earlier. We now describe its two phases.

## 5.1 MERGING PHASE

The merging phase realizes the “forward” half of reversible merging: it constructs a compact representation of all task-specific models using a shared basis. For each layer and for each task vector position (a fixed row in  $A_i^l$  or column in  $B_i^l$ ), we gather the task vectors across all the  $n$  models

as  $\mathbf{X} = [\mathbf{x}_1 \dots \mathbf{x}_n]^\top \in \mathbb{R}^{n \times r}$ . We first compute the mean vector  $\boldsymbol{\mu} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i$ , and then zero-center  $\mathbf{X}$  (i.e.,  $\mathbf{X} \leftarrow [\mathbf{x}_1 - \boldsymbol{\mu} \dots \mathbf{x}_n - \boldsymbol{\mu}]^\top$ ). Next, we perform SVD on the centered matrix  $\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ , and select the top- $p$  right singular vectors to form the optimal orthonormal basis  $\mathbf{W}^*$  minimizing the reconstruction error in equation 5, i.e.,  $\mathbf{W}^* = \mathbf{V}_{[:,1:p]} \in \mathbb{R}^{r \times p}$ . We then compute the corresponding coefficient matrix  $\mathbf{C}^* = \mathbf{X}\mathbf{W}^* \in \mathbb{R}^{n \times p}$ , storing the linear combination weights to reconstruct the original task vectors. The  $i$ -th row of  $\mathbf{C}^*$ ,  $[c_{i1}, \dots, c_{ip}]$ , contains the coefficients needed to reconstruct  $\mathbf{x}_i$ .

By the end of this phase, the original task vectors  $\{\mathbf{x}_i\}_{i=1}^n$  are discarded, and only the basis  $\mathbf{W}^*$ , the coefficients  $\mathbf{C}^*$ , and the mean vector  $\boldsymbol{\mu}$  are retained per task vector position. This delivers the compression at the heart of our efficiency claim: from  $rn$  parameters (to retain all task vectors) down to  $p(r + n) + r$  per position. The phase runs offline, prior to inference.

## 5.2 INFERENCE PHASE

The inference phase realizes the “reverse” half—the reversibility that names our method. Following common practice in model merging, we assume the target task for each input is either specified by the user or determined by an oracle router that assigns the input to the most suitable model index. Once the target index  $i$  is known, we reconstruct its task vector  $\mathbf{x}_i$ , denoted  $\hat{\mathbf{x}}_i$ , from the compact representation  $(\mathbf{W}^*, \mathbf{C}^*, \boldsymbol{\mu})$  stored at each position, and then recover  $\mathbf{A}_i^l$  and  $\mathbf{B}_i^l$  by concatenating the reconstructed vectors  $\hat{\mathbf{x}}_i$  across all positions (the corresponding rows of  $\mathbf{A}_i^l$  or columns of  $\mathbf{B}_i^l$ ).

Recall that for each task vector position,  $\mathbf{W}^* \in \mathbb{R}^{r \times p}$  encodes the orthonormal basis vectors,  $\mathbf{C}^* \in \mathbb{R}^{n \times p}$  stores the task-specific coefficients, and  $\boldsymbol{\mu} \in \mathbb{R}^r$  is the mean vector across all task vectors. The reconstruction is performed by linearly combining the basis directions using task-specific coefficients shifted with the mean vector  $\boldsymbol{\mu}$ , as the following:

$$\hat{\mathbf{x}}_i = \sum_{j=1}^p c_{ij}^* \mathbf{w}_j^* + \boldsymbol{\mu} = \mathbf{C}_{[i,:]}^* \mathbf{W}^{*\top} + \boldsymbol{\mu}.$$

The resulting  $\hat{\mathbf{x}}_i \in \mathbb{R}^r$  approximates the original task vector  $\mathbf{x}_i$  discarded during merging. Applying this reconstruction independently at every position (each row of  $\mathbf{A}_i^l$  and each column of  $\mathbf{B}_i^l$ ) and every layer fully recovers the low-rank delta  $\hat{\Delta}_i$  for task-specific model  $i$ —the “reversion” promised by Contribution 2. Algorithm 1 summarizes both phases. Having specified all three contributions, we now turn to the empirical evaluation that validates them against standard baselines.

## 6 EXPERIMENTS

We now empirically validate our contributions across a variety of compression settings, tasks, and model architectures. The experiments are designed around two questions that directly test the value claimed for RMM (Contribution 3): (i) does RMM preserve the performance of low-rank task-specific models while reducing overall storage cost, and (ii) how does it compare to the conventional merging baselines that Contribution 1 showed to fail? The results below answer both affirmatively.

### 6.1 EXPERIMENTAL SETUP

Our primary evaluation uses eight RoBERTa-base models fine-tuned for eight different tasks (Zhuang et al., 2021). Each model is individually compressed with either PT-SVD or LoRA at target ranks  $r \in \{16, 32, 64, 128\}$ ; we then apply RMM to the resulting low-rank models and compare against the standard merging baselines Task Arithmetic (TA) (Ilharco et al., 2023), TIES-merging (Yadav et al., 2023), and DARE (Yu et al., 2024). To bound the achievable performance, we also report the individual compressed models before merging (“No merging”). Appendix A.2 broadens the evidence for Contribution 3 with additional experiments on OPT-1.3b (Zhang et al., 2022) and on ViT-B/32 (Radford et al., 2021) for vision tasks, using different numbers of tasks.

Following the GLUE benchmark protocol (Wang et al., 2018), we score each task with its standard metric: QNLI (question-answering inference), MRPC (paraphrase detection), SST-2 (sentiment classification), MNLI (multi-genre natural language inference), QQP (duplicate question detection), and RTE (textual entailment) by accuracy; CoLA (linguistic acceptability) by Matthews

---

**Algorithm 1** Reversible Model Merging (RMM)

---

```

1: Merging Phase:
2: Require: Low-rank deltas  $A_i^l$  and  $B_i^l$ , compression rank  $r$ , number of basis vectors  $p$ 
3: for  $l = 1$  to  $L$  do
4:   for each task vector position (each row in  $A_i^l$  or each column in  $B_i^l$ ) do
5:     Gather task vectors  $x_i$  of each model  $i$  ( $1 \leq i \leq n$ )
6:      $X \leftarrow [x_1 \dots x_n]^\top \in \mathbb{R}^{n \times r}$ 
7:      $\mu \leftarrow \sum_{i=1}^n x_i \in \mathbb{R}^r$ 
8:      $X \leftarrow [x_1 - \mu \dots x_n - \mu]^\top$ 
9:      $U, \Sigma, V^\top \leftarrow \text{SVD}(X)$ 
10:     $W^* \leftarrow V_{[:,1:p]} \in \mathbb{R}^{r \times p}$ 
11:     $C^* \leftarrow XW^* \in \mathbb{R}^{n \times p}$ 
12:    Store  $(W^*, C^*, \mu)$  and discard task vectors  $\{x_1 \dots x_n\}$ 
13:   end for
14: end for
15: Inference Phase:
16: Require: target task  $i$ , stored representation  $(W^*, C^*, \mu)$  for each task vector position
17: for  $l = 1$  to  $L$  do
18:   for each task vector position do
19:      $\hat{x}_i = C_{[i,:]}^* W^{*\top} + \mu$ 
20:   end for
21:    $\hat{A}_i^l = \text{Concatenate}(\hat{x}_i \text{ vectors across all row positions})$ 
22:    $\hat{B}_i^l = \text{Concatenate}(\hat{x}_i \text{ vectors across all column positions})$ 
23: end for
24:  $\hat{\Delta}_i = \{\hat{A}_i^l \hat{B}_i^l\}_{l=1}^L$ 
25: Inference with the reconstructed model  $\theta_{pre} + \hat{\Delta}_i$ 

```

---

correlation; and STS-B (semantic textual similarity) by Pearson correlation. Overall performance is the average across tasks. To make the storage side of the trade-off explicit, we also report each method’s relative storage cost—the fraction of parameters retained relative to storing all individual task-specific adapters. Baselines that keep a single merged model have a fixed cost of  $\frac{r}{rn} = \frac{1}{n}$ , whereas RMM with  $p$  basis components costs  $\frac{p(r+n)+r}{rn}$ , as derived in Section 5.1. Reporting both accuracy and storage lets the tables speak directly to Contribution 3’s tunable trade-off.

## 6.2 RESULTS AND DISCUSSION

Tables 1 and 2 report RMM against the baselines under PT-SVD and LoRA compression. The headline finding, supporting Contribution 3, is unambiguous: across both compression schemes and all four ranks  $r \in \{16, 32, 64, 128\}$ , RMM outperforms TA, TIES, and DARE by a large margin. At lower ranks ( $r = 16, 32$ ), the baselines fail catastrophically—only 31–42% average score, and near-zero on structurally sensitive tasks such as CoLA and STS-B—exactly the collapse anticipated by Contribution 1. RMM instead recovers much of the lost performance:  $p = 2$  already yields substantial gains (e.g., 57.24% under PT-SVD and 51.41% under LoRA at  $r = 16$ ), and  $p = 3$  pushes further (up to 72.22% and 58.64% at  $r = 16$ ), sharply narrowing the gap to the individual compressed models. At higher ranks ( $r = 64, 128$ ), where compression error shrinks, RMM continues to dominate: with  $p = 3$  it reaches 64.34% and 60.58% under PT-SVD and 55.40% and 61.35% under LoRA, while the single-model baselines remain stuck near 40%. Even  $p = 2$  retains a clear edge, confirming that the reversible formulation of Contribution 2 sidesteps the destructive interference that cripples prior methods.

One trend is worth noting: RMM’s relative gain over the baselines shrinks as the rank grows. A plausible explanation is that higher-rank task representations occupy higher-dimensional, more expressive, and less aligned spaces, which are harder to compress into a shared low-dimensional basis. This nuance does not weaken the central claim. Taken together, the results establish Contribution 3’s core value proposition: RMM delivers a consistent advantage across both compression regimes and all ranks, preserving high task performance in exchange for some extra storage, with the number

| $r$ | Method          | Storage | QNLI  | MRPC  | SST-2 | MNLI  | QQP   | RTE   | COLA  | STS   | Average |
|-----|-----------------|---------|-------|-------|-------|-------|-------|-------|-------|-------|---------|
| 16  | No merging      | 100%    | 92.02 | 85.68 | 93.69 | 85.21 | 87.19 | 77.98 | 58.05 | 90.59 | 83.80   |
|     | TA              | 13%     | 50.59 | 33.51 | 49.08 | 31.82 | 36.82 | 47.29 | 0.00  | 4.19  | 31.66   |
|     | TIES            | 13%     | 50.56 | 34.78 | 49.08 | 31.82 | 36.82 | 47.29 | 0.00  | 3.75  | 31.76   |
|     | DARE            | 13%     | 50.80 | 33.51 | 49.08 | 31.82 | 36.82 | 47.29 | 0.00  | 5.07  | 31.80   |
|     | RMM ( $p = 2$ ) | 50%     | 51.20 | 78.20 | 88.30 | 66.74 | 86.02 | 47.29 | 6.56  | 33.59 | 57.24   |
|     | RMM ( $p = 3$ ) | 69%     | 88.83 | 86.26 | 92.89 | 79.58 | 87.18 | 48.74 | 33.85 | 60.46 | 72.22   |
| 32  | No merging      | 100%    | 92.31 | 85.86 | 94.15 | 86.38 | 88.62 | 78.34 | 59.33 | 90.68 | 84.46   |
|     | TA              | 13%     | 50.59 | 33.51 | 49.08 | 31.82 | 36.82 | 47.29 | 0.00  | 4.21  | 31.66   |
|     | TIES            | 13%     | 50.56 | 35.19 | 49.08 | 31.82 | 36.82 | 47.29 | 0.00  | 3.87  | 31.83   |
|     | DARE            | 13%     | 50.72 | 33.51 | 49.08 | 31.82 | 36.82 | 47.29 | 0.00  | 3.48  | 31.59   |
|     | RMM ( $p = 2$ ) | 44%     | 50.54 | 74.32 | 55.05 | 62.68 | 87.07 | 47.29 | 0.00  | 23.18 | 50.02   |
|     | RMM ( $p = 3$ ) | 59%     | 84.66 | 86.32 | 92.89 | 79.70 | 88.05 | 47.29 | 23.57 | 44.54 | 68.38   |
| 64  | No merging      | 100%    | 92.51 | 86.03 | 94.38 | 87.18 | 90.15 | 78.70 | 61.11 | 90.72 | 85.10   |
|     | TA              | 13%     | 50.61 | 33.51 | 49.08 | 31.82 | 36.82 | 47.29 | 0.00  | 4.23  | 31.67   |
|     | TIES            | 13%     | 50.58 | 35.13 | 49.08 | 31.82 | 36.82 | 47.29 | 0.00  | 3.80  | 31.82   |
|     | DARE            | 13%     | 50.58 | 33.51 | 49.08 | 31.82 | 36.82 | 47.29 | 0.00  | 5.22  | 31.79   |
|     | RMM ( $p = 2$ ) | 41%     | 50.53 | 73.10 | 49.08 | 58.48 | 86.98 | 47.29 | 0.00  | 17.50 | 47.87   |
|     | RMM ( $p = 3$ ) | 55%     | 74.14 | 86.20 | 92.43 | 78.83 | 89.00 | 47.29 | 13.16 | 33.70 | 64.34   |
| 128 | No merging      | 100%    | 92.68 | 86.14 | 94.50 | 87.33 | 91.05 | 79.78 | 60.16 | 90.74 | 85.30   |
|     | TA              | 13%     | 50.59 | 33.51 | 49.08 | 31.82 | 36.82 | 47.29 | 0.00  | 4.23  | 31.67   |
|     | TIES            | 13%     | 50.58 | 35.54 | 49.08 | 31.82 | 36.82 | 47.29 | 0.00  | 3.93  | 31.88   |
|     | DARE            | 13%     | 50.61 | 34.78 | 49.08 | 31.82 | 36.82 | 47.29 | 0.00  | 2.41  | 31.60   |
|     | RMM ( $p = 2$ ) | 39%     | 50.54 | 70.78 | 49.08 | 54.69 | 84.68 | 47.29 | 0.00  | 14.53 | 46.45   |
|     | RMM ( $p = 3$ ) | 52%     | 59.20 | 85.80 | 92.32 | 77.50 | 89.02 | 47.29 | 6.56  | 26.94 | 60.58   |

Table 1: Main evidence for Contribution 3 (PT-SVD): GLUE performance when merging eight RoBERTa-base models compressed with PT-SVD at ranks  $r = 16, 32, 64, 128$ . Both RMM variants outperform every baseline at every rank, at the cost of a tunable increase in offline storage.

| $r$ | Method          | Storage | QNLI  | MRPC  | SST-2 | MNLI  | QQP   | RTE   | COLA  | STS   | Average |
|-----|-----------------|---------|-------|-------|-------|-------|-------|-------|-------|-------|---------|
| 16  | No merging      | 100%    | 61.38 | 83.48 | 94.50 | 83.90 | 87.56 | 68.23 | 53.92 | 88.71 | 77.71   |
|     | TA              | 13%     | 50.59 | 33.51 | 72.13 | 34.79 | 64.46 | 47.29 | 0.00  | 21.81 | 40.57   |
|     | TIES            | 13%     | 50.63 | 33.68 | 74.31 | 35.64 | 67.45 | 47.29 | 0.00  | 26.38 | 41.92   |
|     | DARE            | 13%     | 50.63 | 33.51 | 73.74 | 35.97 | 65.35 | 47.29 | 0.00  | 21.62 | 41.01   |
|     | RMM ( $p = 2$ ) | 50%     | 51.13 | 34.14 | 90.71 | 60.48 | 84.41 | 47.29 | 0.00  | 43.14 | 51.41   |
|     | RMM ( $p = 3$ ) | 69%     | 55.92 | 38.14 | 93.12 | 82.95 | 87.30 | 47.29 | 6.56  | 57.81 | 58.64   |
| 32  | No merging      | 100%    | 60.13 | 81.28 | 93.81 | 84.46 | 88.18 | 62.82 | 53.12 | 88.53 | 76.54   |
|     | TA              | 13%     | 50.91 | 68.06 | 54.70 | 33.77 | 65.50 | 55.96 | 0.00  | -0.48 | 41.05   |
|     | TIES            | 13%     | 51.03 | 68.06 | 56.77 | 34.24 | 66.27 | 58.12 | 0.00  | 1.02  | 41.94   |
|     | DARE            | 13%     | 50.96 | 64.64 | 63.07 | 35.12 | 63.66 | 55.60 | 0.00  | 1.01  | 41.76   |
|     | RMM ( $p = 2$ ) | 44%     | 52.08 | 65.39 | 81.54 | 62.46 | 77.06 | 57.40 | 0.00  | 9.95  | 50.73   |
|     | RMM ( $p = 3$ ) | 59%     | 54.62 | 69.97 | 92.20 | 81.29 | 85.97 | 57.40 | 4.64  | 24.08 | 58.77   |
| 64  | No merging      | 100%    | 59.62 | 84.12 | 94.15 | 85.20 | 88.75 | 67.15 | 58.31 | 88.70 | 78.25   |
|     | TA              | 13%     | 54.75 | 68.52 | 77.29 | 37.24 | 66.82 | 47.29 | 0.00  | -3.53 | 39.80   |
|     | TIES            | 13%     | 54.82 | 68.75 | 80.73 | 37.67 | 66.82 | 47.29 | 0.00  | -2.48 | 40.45   |
|     | DARE            | 13%     | 53.71 | 68.58 | 78.56 | 38.33 | 66.82 | 47.29 | 0.00  | -4.28 | 39.88   |
|     | RMM ( $p = 2$ ) | 41%     | 55.74 | 69.28 | 87.27 | 42.81 | 77.72 | 47.29 | 0.00  | 4.91  | 48.13   |
|     | RMM ( $p = 3$ ) | 55%     | 59.07 | 70.14 | 90.83 | 79.11 | 82.06 | 47.29 | 0.00  | 14.70 | 55.40   |
| 128 | No merging      | 100%    | 61.58 | 85.04 | 93.58 | 85.71 | 89.68 | 57.04 | 59.35 | 89.68 | 77.71   |
|     | TA              | 13%     | 51.49 | 69.22 | 54.47 | 37.16 | 63.27 | 47.29 | 0.00  | -1.74 | 40.15   |
|     | TIES            | 13%     | 51.91 | 68.93 | 57.34 | 37.81 | 63.36 | 47.29 | 0.00  | -0.25 | 40.80   |
|     | DARE            | 13%     | 51.97 | 68.99 | 58.60 | 35.64 | 63.33 | 47.29 | 0.00  | -0.44 | 40.67   |
|     | RMM ( $p = 2$ ) | 39%     | 57.18 | 67.83 | 88.53 | 49.54 | 84.64 | 47.29 | 0.00  | 28.00 | 52.88   |
|     | RMM ( $p = 3$ ) | 52%     | 59.13 | 39.19 | 92.09 | 80.08 | 86.39 | 47.29 | 15.45 | 71.18 | 61.35   |

Table 2: Main evidence for Contribution 3 (LoRA): GLUE performance when merging eight RoBERTa-base models compressed with LoRA at ranks  $r = 16, 32, 64, 128$ . The RMM advantage over TA, TIES, and DARE holds under LoRA as well as PT-SVD (Table 1).

of reversible components acting as a direct accuracy–storage dial. Crucially, that extra cost is **offline memory**—typically CPU-based and inexpensive—whereas task performance governs runtime quality, so trading a modest increase in CPU-side storage for substantially better accuracy is usually worthwhile.

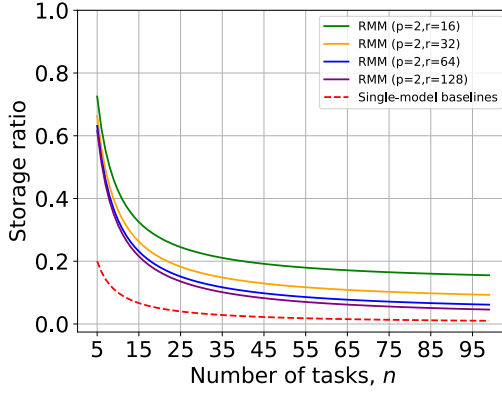


Figure 4: Scalability evidence for RMM ( $p = 2$ ): the storage ratio (memory required relative to storing all models) decreases as the number of tasks grows, illustrating the sublinear growth that underpins RMM’s practicality at scale.

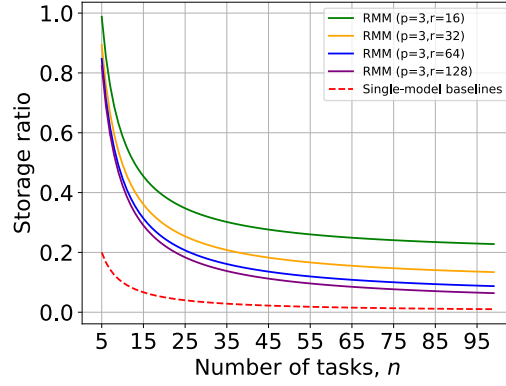


Figure 5: Scalability evidence for RMM ( $p = 3$ ): as with  $p = 2$  (Fig. 4), the storage ratio relative to storing all models falls with the number of tasks, confirming that the storage benefit strengthens as more models are added.

### 6.3 SCALABILITY OF RMM

Beyond raw accuracy, a further benefit of Contribution 3 is how gracefully RMM scales as the number of models  $n$  grows. Figures 4 and 5 plot the storage ratio (relative to storing all individual task-specific models) against the number of tasks for  $p = 2$  and  $p = 3$ . The message is clear: the relative storage cost *decreases* as tasks accumulate. This is because storage grows sublinearly with the number of models—once the basis is fixed, each new task adds only a single row to the coefficient matrix. Traditional approaches offer no such property: they must either store one merged model (sacrificing performance, as Contribution 1 showed) or store every task-specific model independently (scaling linearly). This sublinear scaling is what makes RMM practical in large-scale multi-task, continual, or federated settings that must manage hundreds of models without compromising task performance.

## 7 CONCLUSION

This paper delivered three contributions to model merging in the low-rank regime. First, we diagnosed a failure mode: applying conventional merging methods to low-rank compressed models causes severe, often catastrophic, performance loss, traceable to amplified task interference and the broken coupling between low-rank factors. Second, motivated by this diagnosis, we reframed merging itself—rather than collapsing all task-specific knowledge into a single model, we store a compact set of intermediate representations, far fewer than the original models yet more flexible than a single aggregate, from which each model can be accurately reconstructed. Third, we turned this reformulation into Reversible Model Merging (RMM), a data-free, closed-form method that jointly selects the optimal basis and the task-specific coefficients, and exposes a controllable performance–storage trade-off via a single hyperparameter. Our experiments confirmed all three: RMM consistently and substantially outperforms prior merging methods on compressed models while adhering to strict memory budgets, and its storage advantage strengthens as the number of tasks grows.

More broadly, these contributions move model merging beyond heuristic, one-shot techniques and formalize it as a structured, reversible process. This perspective broadens the design space for efficient multi-task deployment and opens promising directions for future work in performance-aware compression and reconstruction.

## REFERENCES

Mircea Cimpoi, Subhransu Maji, Iasonas Kokkinos, Sammy Mohamed, and Andrea Vedaldi. Describing textures in the wild. In *Proceedings of the IEEE conference on computer vision and*

540 *pattern recognition*, pp. 3606–3613, 2014.

541

542 Christopher Clark, Kenton Lee, Ming-Wei Chang, Tom Kwiatkowski, Michael Collins, and Kristina  
543 Toutanova. BoolQ: Exploring the surprising difficulty of natural yes/no questions. In Jill  
544 Burstein, Christy Doran, and Thamar Solorio (eds.), *Proceedings of the 2019 Conference of*  
545 *the North American Chapter of the Association for Computational Linguistics: Human Lan-*  
546 *guage Technologies, Volume 1 (Long and Short Papers)*, pp. 2924–2936, Minneapolis, Min-  
547 nesota, June 2019. Association for Computational Linguistics. doi: 10.18653/v1/N19-1300. URL  
548 <https://aclanthology.org/N19-1300/>.

549 Antonio Andrea Gargiulo, Donato Crisostomi, Maria Sofia Bucarelli, Simone Scardapane, Fabrizio  
550 Silvestri, and Emanuele Rodola. Task singular vectors: Reducing task interference in model  
551 merging. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pp. 18695–  
552 18705, 2025.

553

554 Patrick Helber, Benjamin Bischke, Andreas Dengel, and Damian Borth. Eurosat: A novel dataset  
555 and deep learning benchmark for land use and land cover classification. *IEEE Journal of Selected*  
556 *Topics in Applied Earth Observations and Remote Sensing*, 12(7):2217–2226, 2019.

557 Roger A. Horn and Charles R. Johnson. *Matrix Analysis*. Cambridge University Press, 2 edition,  
558 2012.

559

560 Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, and  
561 Weizhu Chen. Lora: Low-rank adaptation of large language models. *CoRR*, abs/2106.09685,  
562 2021. URL <https://arxiv.org/abs/2106.09685>.

563

564 Chenyu Huang, Peng Ye, Tao Chen, Tong He, Xiangyu Yue, and Wanli Ouyang. Emr-merging:  
565 Tuning-free high-performance model merging. *Advances in Neural Information Processing Sys-*  
566 *tems*, 37:122741–122769, 2024.

567 Gabriel Ilharco, Marco Tulio Ribeiro, Mitchell Wortsman, Ludwig Schmidt, Hannaneh Hajishirzi,  
568 and Ali Farhadi. Editing models with task arithmetic. In *The Eleventh International Confer-*  
569 *ence on Learning Representations*, 2023. URL [https://openreview.net/forum?id=](https://openreview.net/forum?id=6t0Kwf8-jrj)  
570 [6t0Kwf8-jrj](https://openreview.net/forum?id=6t0Kwf8-jrj).

571

572 Xisen Jin, Xiang Ren, Daniel Preotiuc-Pietro, and Pengxiang Cheng. Dataless knowledge fusion  
573 by merging weights of language models. In *The Eleventh International Conference on Learning*  
574 *Representations*, 2023. URL [https://openreview.net/forum?id=](https://openreview.net/forum?id=FCnohuR6AnM)  
575 [FCnohuR6AnM](https://openreview.net/forum?id=FCnohuR6AnM).

576

577 Jonathan Krause, Michael Stark, Jia Deng, and Li Fei-Fei. 3d object representations for fine-grained  
578 categorization. In *2013 IEEE International Conference on Computer Vision Workshops*, pp. 554–  
579 561, 2013. doi: 10.1109/ICCVW.2013.77.

580

581 Yann LeCun, Léon Bottou, Yoshua Bengio, and Patrick Haffner. Gradient-based learning applied to  
582 document recognition. *Proceedings of the IEEE*, 86(11):2278–2324, 1998.

583

584 Michael S Matena and Colin A Raffel. Merging models with fisher-weighted averaging. *Advances*  
585 *in Neural Information Processing Systems*, 35:17703–17716, 2022.

586

587 Yuval Netzer, Tao Wang, Adam Coates, Alessandro Bissacco, Bo Wu, and Andrew Y. Ng. Reading  
588 digits in natural images with unsupervised feature learning. In *NIPS Workshop on Deep Learning*  
589 *and Unsupervised Feature Learning 2011*, 2011. URL [http://ufldl.stanford.edu/](http://ufldl.stanford.edu/housenumbers/nips2011_housenumbers.pdf)  
590 [housenumbers/nips2011\\_housenumbers.pdf](http://ufldl.stanford.edu/housenumbers/nips2011_housenumbers.pdf).

591

592 Bowen Ping, Shuo Wang, Hanqing Wang, Xu Han, Yuzhuang Xu, Yukun Yan, Yun Chen,  
593 Baobao Chang, Zhiyuan Liu, and Maosong Sun. Delta-come: Training-free delta-compression  
with mixed-precision for large language models. In A. Globerson, L. Mackey, D. Bel-  
grave, A. Fan, U. Paquet, J. Tomczak, and C. Zhang (eds.), *Advances in Neural In-*  
formation Processing Systems, volume 37, pp. 31056–31077. Curran Associates, Inc.,  
2024. URL [https://proceedings.neurips.cc/paper\\_files/paper/2024/](https://proceedings.neurips.cc/paper_files/paper/2024/file/37664246a1e07e212ddacea6e5a523f2-Paper-Conference.pdf)  
[file/37664246a1e07e212ddacea6e5a523f2-Paper-Conference.pdf](https://proceedings.neurips.cc/paper_files/paper/2024/file/37664246a1e07e212ddacea6e5a523f2-Paper-Conference.pdf).

- Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In Marina Meila and Tong Zhang (eds.), *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pp. 8748–8763. PMLR, 18–24 Jul 2021. URL <https://proceedings.mlr.press/v139/radford21a.html>.
- Simo Ryu, Seunghyun Seo, and Jaeyun Yoo. Efficient storage of fine-tuned models via low-rank approximation of weight residuals. *arXiv preprint arXiv:2305.18425*, 2023.
- Johannes Stallkamp, Marc Schlipsing, Jan Salmen, and Christian Igel. The german traffic sign recognition benchmark: A multi-class classification competition. In *The 2011 International Joint Conference on Neural Networks*, pp. 1453–1460, 2011. doi: 10.1109/IJCNN.2011.6033395.
- Dennis Tang, Prateek Yadav, Yi-Lin Sung, Jaehong Yoon, and Mohit Bansal. LoRA merging with SVD: Understanding interference and preserving performance. In *ICML 2025 Workshop on Reliable and Responsible Foundation Models*, 2025. URL <https://openreview.net/forum?id=t9FrMvITaP>.
- Alex Wang, Amanpreet Singh, Julian Michael, Felix Hill, Omer Levy, and Samuel Bowman. GLUE: A multi-task benchmark and analysis platform for natural language understanding. In Tal Linzen, Grzegorz Chrupała, and Afra Alishahi (eds.), *Proceedings of the 2018 EMNLP Workshop Black-boxNLP: Analyzing and Interpreting Neural Networks for NLP*, pp. 353–355, Brussels, Belgium, November 2018. Association for Computational Linguistics. doi: 10.18653/v1/W18-5446. URL <https://aclanthology.org/W18-5446/>.
- Xin Wang, Samiul Alam, Zhongwei Wan, Hui Shen, and Mi Zhang. SVD-LLM v2: Optimizing singular value truncation for large language model compression. In Luis Chiruzzo, Alan Ritter, and Lu Wang (eds.), *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pp. 4287–4296, Albuquerque, New Mexico, April 2025a. Association for Computational Linguistics. ISBN 979-8-89176-189-6. doi: 10.18653/v1/2025.naacl-long.217. URL <https://aclanthology.org/2025.naacl-long.217/>.
- Xin Wang, Yu Zheng, Zhongwei Wan, and Mi Zhang. SVD-LLM: Truncation-aware singular value decomposition for large language model compression. In *The Thirteenth International Conference on Learning Representations*, 2025b. URL <https://openreview.net/forum?id=LNyIUouhdt>.
- Yongxian Wei, Anke Tang, Li Shen, Zixuan Hu, Chun Yuan, and Xiaochun Cao. Modeling multi-task model merging as adaptive projective gradient descent. In *Forty-second International Conference on Machine Learning*, 2025. URL <https://openreview.net/forum?id=EgQKRSR5Pa>.
- Mitchell Wortsman, Gabriel Ilharco, Samir Ya Gadre, Rebecca Roelofs, Raphael Gontijo-Lopes, Ari S Morcos, Hongseok Namkoong, Ali Farhadi, Yair Carmon, Simon Kornblith, et al. Model soups: averaging weights of multiple fine-tuned models improves accuracy without increasing inference time. In *International conference on machine learning*, pp. 23965–23998. PMLR, 2022.
- Jianxiong Xiao, James Hays, Krista A. Ehinger, Aude Oliva, and Antonio Torralba. Sun database: Large-scale scene recognition from abbey to zoo. In *2010 IEEE Computer Society Conference on Computer Vision and Pattern Recognition*, pp. 3485–3492, 2010. doi: 10.1109/CVPR.2010.5539970.
- Prateek Yadav, Derek Tam, Leshem Choshen, Colin Raffel, and Mohit Bansal. TIES-merging: Resolving interference when merging models. In *Thirty-seventh Conference on Neural Information Processing Systems*, 2023. URL <https://openreview.net/forum?id=xtaX3WycJl>.
- Enneng Yang, Zhenyi Wang, Li Shen, Shiwei Liu, Guibing Guo, Xingwei Wang, and Dacheng Tao. Adamerging: Adaptive model merging for multi-task learning. In *The Twelfth International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=nZP6NgD3QY>.

- Le Yu, Bowen Yu, Haiyang Yu, Fei Huang, and Yongbin Li. Language models are super mario: Absorbing abilities from homologous models as a free lunch. In *Forty-first International Conference on Machine Learning*, 2024.
- Juzheng Zhang, Jiacheng You, Ashwinee Panda, and Tom Goldstein. LoRI: Reducing cross-task interference in multi-task low-rank adaptation. In *Second Conference on Language Modeling*, 2025. URL <https://openreview.net/forum?id=b8cW86QcOD>.
- Susan Zhang, Stephen Roller, Naman Goyal, Mikel Artetxe, Moya Chen, Shuohui Chen, Christopher Dewan, Mona Diab, Xian Li, Xi Victoria Lin, et al. Opt: Open pre-trained transformer language models. *arXiv preprint arXiv:2205.01068*, 2022.
- Ziyu Zhao, Tao Shen, Didi Zhu, Zexi Li, Jing Su, Xuwu Wang, and Fei Wu. Merging loRAs like playing LEGO: Pushing the modularity of loRA to extremes through rank-wise clustering. In *The Thirteenth International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=j6fsbpAllN>.
- Shenghe Zheng, Hongzhi Wang, Chenyu Huang, Xiaohui Wang, Tao Chen, Jiayuan Fan, Shuyue Hu, and Peng Ye. Decouple and orthogonalize: A data-free framework for lora merging. *arXiv preprint arXiv:2505.15875*, 2025.
- Liu Zhuang, Lin Wayne, Shi Ya, and Zhao Jun. A robustly optimized BERT pre-training approach with post-training. In Sheng Li, Maosong Sun, Yang Liu, Hua Wu, Kang Liu, Wanxiang Che, Shizhu He, and Gaoqi Rao (eds.), *Proceedings of the 20th Chinese National Conference on Computational Linguistics*, pp. 1218–1227, Huhhot, China, August 2021. Chinese Information Processing Society of China. URL <https://aclanthology.org/2021.ccl-1.108/>.

## A APPENDIX

### A.1 PROOF OF THEOREM 1

Let  $\mathbf{A} = \mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top$ . By definition of the Frobenius norm,  $\|\mathbf{A}\|_F^2 = \text{Tr}(\mathbf{A}^\top \mathbf{A})$ , where  $\text{Tr}(\cdot)$  denotes trace of a square matrix. Expanding gives:

$$\|\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top\|_F^2 = \text{Tr}((\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top)^\top (\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top)).$$

We expand the term in  $\text{Tr}(\cdot)$ :

$$\begin{aligned} (\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top)^\top (\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top) &= (\mathbf{X}^\top - \mathbf{W}\mathbf{W}^\top \mathbf{X}^\top)(\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top) \\ &= \mathbf{X}^\top \mathbf{X} - \mathbf{X}^\top \mathbf{X}\mathbf{W}\mathbf{W}^\top - \mathbf{W}\mathbf{W}^\top \mathbf{X}^\top \mathbf{X} \\ &\quad + \mathbf{W}\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}\mathbf{W}^\top. \end{aligned}$$

Next, we take the trace of the expanded form. Using cyclicity of trace and  $\mathbf{W}^\top \mathbf{W} = \mathbf{I}_p$ :

$$\begin{aligned} \|\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top\|_F^2 &= \text{Tr}(\mathbf{X}^\top \mathbf{X}) - 2\text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}) + \text{Tr}(\mathbf{W}^\top \mathbf{W}\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}) \\ &= \text{Tr}(\mathbf{X}^\top \mathbf{X}) - 2\text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}) + \text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}) \\ &= \text{Tr}(\mathbf{X}^\top \mathbf{X}) - \text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}). \end{aligned}$$

Since  $\text{Tr}(\mathbf{X}^\top \mathbf{X})$  is a constant, minimizing  $\|\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top\|_F^2$  is equivalent to maximizing  $\text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W})$ . Hence,

$$\min_{\mathbf{W}^\top \mathbf{W} = \mathbf{I}_p} \|\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top\|_F^2 \iff \max_{\mathbf{W}^\top \mathbf{W} = \mathbf{I}_p} \text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}).$$

By the Rayleigh quotient theorem (Horn & Johnson, 2012), the maximizer takes the columns of  $\mathbf{W}$  to be the top- $p$  eigenvectors of  $\mathbf{X}^\top \mathbf{X}$ . This is exactly the principal component analysis (PCA) solution: it identifies the  $p$ -dimensional subspace that maximizes projected variance and, equivalently, minimizes squared reconstruction error—the very objective that formalizes reversibility in Contribution 3. Moreover, if  $\mathbf{X} = \mathbf{U}\Sigma\mathbf{V}^\top$  is the SVD of  $\mathbf{X}$ , then  $\mathbf{X}^\top \mathbf{X} = \mathbf{V}\Sigma^2\mathbf{V}^\top$ , so the eigenvectors of  $\mathbf{X}^\top \mathbf{X}$  coincide with the right singular vectors (columns of  $\mathbf{V}$ ). Hence the same solution follows from the top- $p$  right singular vectors of  $\mathbf{X}$ , which is numerically more stable and often computationally preferable to forming  $\mathbf{X}^\top \mathbf{X}$  directly, especially when  $r$  is large. This closed form is what makes RMM data-free and cheap to compute.

| <b>r</b> | <b>Method</b>          | <b>Storage</b> | <b>QNLI</b> | <b>MRPC</b> | <b>SST-2</b> | <b>MNLI</b> | <b>QQP</b> | <b>BOOLQ</b> | <b>Average</b> |
|----------|------------------------|----------------|-------------|-------------|--------------|-------------|------------|--------------|----------------|
| 16       | No merging             | 100%           | 91.74       | 71.54       | 95.30        | 84.19       | 88.14      | 71.47        | 83.73          |
|          | Task-Arithmetic        | 17%            | 50.34       | 37.16       | 58.49        | 34.43       | 63.30      | 41.87        | 47.60          |
|          | TIES                   | 17%            | 49.79       | 36.75       | 61.24        | 35.57       | 63.55      | 40.67        | 47.93          |
|          | DARE                   | 17%            | 49.95       | 37.68       | 56.08        | 34.51       | 63.39      | 42.78        | 47.40          |
|          | <b>RMM</b> ( $p = 2$ ) | 63%            | 88.12       | 44.00       | 93.58        | 73.06       | 85.63      | 59.69        | 74.01          |
|          | <b>RMM</b> ( $p = 3$ ) | 85%            | 91.58       | 49.04       | 95.64        | 80.31       | 87.52      | 65.50        | 78.27          |
| 32       | No merging             | 100%           | 92.06       | 72.17       | 95.41        | 84.60       | 89.17      | 72.45        | 84.31          |
|          | Task-Arithmetic        | 17%            | 50.28       | 37.22       | 58.94        | 34.57       | 63.33      | 41.93        | 47.71          |
|          | TIES                   | 17%            | 49.77       | 36.93       | 61.93        | 35.69       | 63.65      | 40.73        | 48.12          |
|          | DARE                   | 17%            | 50.39       | 40.87       | 56.19        | 34.96       | 63.59      | 43.33        | 48.22          |
|          | <b>RMM</b> ( $p = 2$ ) | 56%            | 86.66       | 41.91       | 91.74        | 71.75       | 85.68      | 56.39        | 72.36          |
|          | <b>RMM</b> ( $p = 3$ ) | 76%            | 91.85       | 44.81       | 94.84        | 80.21       | 88.43      | 63.52        | 77.28          |
| 64       | No merging             | 100%           | 92.26       | 72.12       | 95.41        | 84.64       | 90.07      | 72.84        | 84.56          |
|          | Task-Arithmetic        | 17%            | 50.28       | 37.28       | 59.06        | 34.56       | 63.34      | 42.08        | 47.77          |
|          | TIES                   | 17%            | 49.75       | 37.04       | 61.81        | 35.84       | 63.74      | 40.76        | 48.16          |
|          | DARE                   | 17%            | 50.16       | 40.52       | 58.83        | 35.59       | 63.35      | 42.54        | 48.50          |
|          | <b>RMM</b> ( $p = 2$ ) | 53%            | 83.87       | 42.72       | 89.45        | 69.76       | 85.70      | 53.43        | 70.82          |
|          | <b>RMM</b> ( $p = 3$ ) | 71%            | 92.07       | 41.97       | 94.50        | 79.70       | 88.73      | 61.96        | 76.49          |
| 128      | No merging             | 100%           | 92.35       | 72.41       | 95.41        | 84.67       | 90.93      | 73.09        | 84.81          |
|          | Task-Arithmetic        | 17%            | 50.32       | 37.39       | 58.94        | 34.63       | 63.37      | 42.08        | 47.79          |
|          | TIES                   | 17%            | 49.79       | 37.04       | 62.04        | 35.99       | 63.74      | 40.55        | 48.19          |
|          | DARE                   | 17%            | 50.03       | 38.14       | 57.91        | 34.72       | 63.31      | 42.11        | 47.70          |
|          | <b>RMM</b> ( $p = 2$ ) | 52%            | 81.38       | 42.78       | 84.75        | 67.66       | 83.22      | 51.99        | 68.63          |
|          | <b>RMM</b> ( $p = 3$ ) | 69%            | 92.28       | 41.80       | 93.69        | 79.42       | 88.76      | 59.69        | 75.94          |

Table 3: Additional evidence for Contribution 3 (OPT-1.3b): GLUE performance when merging six OPT-1.3b models compressed with PT-SVD at ranks  $r = 16, 32, 64, 128$ . RMM’s advantage over the baselines carries over to a larger, decoder-style language model.

| <b>r</b> | <b>Method</b>          | <b>Storage</b> | <b>MNIST</b> | <b>GTSRB</b> | <b>SVHN</b> | <b>Cars</b> | <b>EuroSAT</b> | <b>DTD</b> | <b>SUN397</b> | <b>Average</b> |
|----------|------------------------|----------------|--------------|--------------|-------------|-------------|----------------|------------|---------------|----------------|
| 16       | No merging             | 100%           | 99.61        | 97.22        | 97.06       | 69.20       | 99.67          | 75.21      | 72.72         | 87.24          |
|          | Task-Arithmetic        | 14%            | 47.19        | 28.29        | 38.64       | 56.27       | 57.52          | 41.38      | 62.92         | 47.46          |
|          | TIES                   | 14%            | 60.14        | 34.52        | 49.03       | 58.55       | 61.70          | 43.67      | 62.87         | 52.93          |
|          | DARE                   | 14%            | 53.80        | 31.00        | 44.96       | 57.60       | 56.93          | 43.99      | 62.42         | 50.10          |
|          | <b>RMM</b> ( $p = 2$ ) | 55%            | 99.16        | 88.52        | 94.89       | 66.61       | 97.44          | 60.64      | 67.86         | 82.16          |
|          | <b>RMM</b> ( $p = 3$ ) | 76%            | 99.58        | 95.65        | 96.54       | 68.30       | 98.93          | 67.18      | 69.50         | 85.10          |
| 32       | No merging             | 100%           | 99.65        | 98.45        | 97.35       | 73.04       | 99.78          | 78.72      | 75.52         | 88.93          |
|          | Task-Arithmetic        | 14%            | 47.90        | 28.31        | 38.68       | 56.24       | 57.52          | 41.38      | 63.00         | 47.58          |
|          | TIES                   | 14%            | 61.25        | 35.28        | 49.95       | 58.80       | 62.41          | 44.04      | 63.05         | 53.54          |
|          | DARE                   | 14%            | 53.05        | 31.04        | 46.34       | 58.54       | 57.74          | 42.87      | 62.87         | 50.35          |
|          | <b>RMM</b> ( $p = 2$ ) | 49%            | 98.98        | 88.23        | 94.29       | 68.32       | 96.81          | 61.91      | 68.36         | 82.41          |
|          | <b>RMM</b> ( $p = 3$ ) | 67%            | 99.50        | 96.53        | 96.68       | 70.90       | 98.93          | 70.43      | 71.23         | 86.31          |
| 64       | No merging             | 100%           | 99.67        | 98.68        | 97.41       | 75.76       | 99.85          | 79.15      | 77.73         | 89.75          |
|          | Task-Arithmetic        | 14%            | 47.99        | 28.30        | 38.71       | 56.21       | 57.59          | 41.38      | 63.22         | 47.63          |
|          | TIES                   | 14%            | 61.83        | 35.93        | 50.28       | 59.08       | 63.07          | 44.15      | 63.17         | 53.93          |
|          | DARE                   | 14%            | 54.78        | 32.79        | 45.05       | 58.23       | 61.30          | 42.87      | 62.77         | 51.11          |
|          | <b>RMM</b> ( $p = 2$ ) | 46%            | 98.46        | 84.43        | 92.83       | 69.36       | 95.59          | 62.39      | 69.77         | 81.83          |
|          | <b>RMM</b> ( $p = 3$ ) | 62%            | 99.45        | 96.14        | 96.33       | 72.84       | 98.56          | 72.07      | 73.85         | 87.03          |
| 128      | No merging             | 100%           | 99.68        | 98.75        | 97.46       | 76.99       | 99.85          | 79.04      | 78.59         | 90.05          |
|          | Task-Arithmetic        | 14%            | 48.00        | 28.31        | 38.73       | 56.21       | 57.63          | 41.44      | 63.35         | 47.67          |
|          | TIES                   | 14%            | 62.08        | 36.36        | 50.48       | 59.23       | 63.41          | 44.68      | 63.25         | 54.21          |
|          | DARE                   | 14%            | 55.16        | 32.07        | 45.47       | 58.55       | 61.85          | 43.09      | 62.85         | 51.29          |
|          | <b>RMM</b> ( $p = 2$ ) | 44%            | 97.22        | 78.56        | 90.63       | 70.22       | 94.00          | 61.49      | 71.03         | 80.45          |
|          | <b>RMM</b> ( $p = 3$ ) | 59%            | 99.28        | 94.59        | 95.95       | 74.24       | 98.19          | 71.86      | 76.02         | 87.16          |

Table 4: Additional evidence for Contribution 3 (vision): performance on vision tasks when merging seven ViT-B/32 models compressed with PT-SVD at ranks  $r = 16, 32, 64, 128$ . RMM extends beyond language to the vision modality, nearly matching the individual models at  $p = 3$ .

## A.2 ADDITIONAL RESULTS

To show that Contribution 3 generalizes beyond RoBERTa, we present extended results for two further pre-trained models spanning both modality and architecture. For **OPT-1.3b**, we evaluate six natural language understanding tasks drawn from GLUE and related benchmarks: QNLI, MRPC, SST-2, MNLI, QQP, and BoolQ (Clark et al., 2019). For **ViT-B/32**, we turn to vision classification over seven diverse datasets: MNIST (LeCun et al., 1998), GTSRB (Stallkamp et al., 2011), SVHN

(Netzer et al., 2011), Cars (Krause et al., 2013), EuroSAT (Helber et al., 2019), DTD (Cimpoi et al., 2014), and SUN397 (Xiao et al., 2010).

The conclusion is the same as in the main experiments. On the six OPT-1.3b models compressed with PT-SVD (Table 3), both RMM variants consistently beat the baselines, and  $p = 3$  narrows the gap to the original individual models while storing substantially fewer parameters. The benefit is even more pronounced on the seven ViT-B/32 models (Table 4): with  $p = 3$ , RMM lands less than 3% below the individual models in average accuracy while cutting storage to just 59% of that needed to retain all models at  $r = 128$ . Together, these results reinforce the central claim—RMM faithfully reconstructs compressed models across languages, architectures, and modalities while remaining storage-efficient.

### A.3 RESULTS FOR 5 TASKS

This section provides the per-task numbers behind Contribution 1’s failure diagnosis. For the experiments in Section 4, we used RoBERTa-base as the pretrained model and evaluated the merging baselines on five GLUE tasks (QNLI, MRPC, SST-2, MNLI, QQP), summarized visually in Figures 1 and 2. For completeness, Table 5 tabulates the LoRA results behind Figure 1—together with the improvement RMM already provides in this five-task setting—and Table 6 tabulates the PT-SVD results behind Figure 2.

| r   | Method                          | Storage | QNLI  | MRPC  | SST-2 | MNLI  | QQP   | Average |
|-----|---------------------------------|---------|-------|-------|-------|-------|-------|---------|
| 16  | No merging                      | 100%    | 61.38 | 83.48 | 94.50 | 83.90 | 87.56 | 82.16   |
|     | Task-Arithmetic                 | 20%     | 50.61 | 33.51 | 73.85 | 34.23 | 64.58 | 51.36   |
|     | TIES                            | 20%     | 50.65 | 33.51 | 75.57 | 35.83 | 66.97 | 52.51   |
|     | DARE                            | 20%     | 50.61 | 33.51 | 79.36 | 34.70 | 63.40 | 52.32   |
|     | <b>RMM (<math>p = 2</math>)</b> | 73%     | 59.64 | 68.64 | 93.00 | 82.68 | 86.89 | 78.17   |
| 32  | No merging                      | 100%    | 60.13 | 81.28 | 93.81 | 84.46 | 88.18 | 81.57   |
|     | Task-Arithmetic                 | 20%     | 51.18 | 67.30 | 58.83 | 34.33 | 68.56 | 56.04   |
|     | TIES                            | 20%     | 51.36 | 68.17 | 59.52 | 34.76 | 71.46 | 57.05   |
|     | DARE                            | 20%     | 50.83 | 64.87 | 68.12 | 35.54 | 68.93 | 57.66   |
|     | <b>RMM (<math>p = 2</math>)</b> | 66%     | 57.72 | 67.54 | 92.55 | 81.80 | 86.13 | 77.15   |
| 64  | No merging                      | 100%    | 59.62 | 84.12 | 94.15 | 85.20 | 88.75 | 82.37   |
|     | Task-Arithmetic                 | 20%     | 54.74 | 68.93 | 80.28 | 35.70 | 36.82 | 55.29   |
|     | TIES                            | 20%     | 55.26 | 68.93 | 78.78 | 36.76 | 36.82 | 55.31   |
|     | DARE                            | 20%     | 54.71 | 68.81 | 74.54 | 38.72 | 36.82 | 54.72   |
|     | <b>RMM (<math>p = 2</math>)</b> | 63%     | 59.82 | 67.48 | 91.51 | 83.21 | 84.41 | 77.29   |
| 128 | No merging                      | 100%    | 61.58 | 85.04 | 93.58 | 85.71 | 89.68 | 83.12   |
|     | Task-Arithmetic                 | 20%     | 52.52 | 68.70 | 54.82 | 37.83 | 63.46 | 55.47   |
|     | TIES                            | 20%     | 52.55 | 69.04 | 57.80 | 38.91 | 63.72 | 56.40   |
|     | DARE                            | 20%     | 55.04 | 69.62 | 54.47 | 38.92 | 64.12 | 56.43   |
|     | <b>RMM (<math>p = 2</math>)</b> | 62%     | 60.21 | 34.96 | 91.86 | 78.59 | 86.65 | 70.45   |

Table 5: Per-task numbers behind Contribution 1 (LoRA): GLUE performance when merging five RoBERTa-base models compressed with LoRA at ranks  $r = 16, 32, 64, 128$ . The baselines collapse relative to the individual models, while RMM ( $p = 2$ ) already recovers most of the lost performance.

| <b>r</b> | <b>Method</b>          | <b>Storage</b> | <b>QNLI</b> | <b>MRPC</b> | <b>SST-2</b> | <b>MNLI</b> | <b>QQP</b> | <b>Average</b> |
|----------|------------------------|----------------|-------------|-------------|--------------|-------------|------------|----------------|
| 16       | No merging             | 100%           | 92.02       | 85.68       | 93.69        | 85.21       | 87.19      | 88.76          |
|          | Task-Arithmetic        | 20%            | 53.03       | 35.19       | 49.08        | 31.82       | 36.82      | 41.19          |
|          | TIES                   | 20%            | 55.46       | 54.32       | 49.08        | 31.82       | 37.21      | 45.58          |
|          | DARE                   | 20%            | 53.69       | 58.67       | 49.08        | 31.82       | 37.08      | 46.07          |
|          | <b>RMM</b> ( $p = 2$ ) | 73%            | 79.72       | 85.16       | 90.83        | 76.58       | 86.09      | 83.68          |
| 32       | No merging             | 100%           | 92.31       | 85.86       | 94.15        | 86.38       | 88.62      | 89.46          |
|          | Task-Arithmetic        | 20%            | 53.23       | 35.59       | 49.08        | 31.82       | 36.82      | 41.31          |
|          | TIES                   | 20%            | 55.78       | 55.19       | 49.08        | 31.82       | 37.62      | 45.90          |
|          | DARE                   | 20%            | 55.89       | 33.74       | 49.20        | 31.82       | 36.85      | 41.50          |
|          | <b>RMM</b> ( $p = 2$ ) | 66%            | 63.87       | 82.78       | 91.28        | 77.12       | 87.05      | 80.42          |
| 64       | No merging             | 100%           | 92.51       | 86.03       | 94.38        | 87.18       | 90.15      | 90.05          |
|          | Task-Arithmetic        | 20%            | 53.49       | 35.88       | 49.08        | 31.82       | 36.82      | 41.42          |
|          | TIES                   | 20%            | 56.14       | 55.94       | 49.08        | 31.82       | 38.08      | 46.21          |
|          | DARE                   | 20%            | 53.51       | 33.68       | 49.08        | 31.82       | 36.82      | 40.98          |
|          | <b>RMM</b> ( $p = 2$ ) | 63%            | 56.12       | 81.22       | 91.06        | 76.74       | 88.07      | 78.64          |
| 128      | No merging             | 100%           | 92.68       | 86.14       | 94.50        | 87.33       | 91.05      | 90.34          |
|          | Task-Arithmetic        | 20%            | 53.60       | 36.06       | 49.08        | 31.82       | 36.82      | 41.48          |
|          | TIES                   | 20%            | 56.40       | 56.35       | 49.08        | 31.82       | 38.61      | 46.45          |
|          | DARE                   | 20%            | 50.52       | 34.96       | 49.08        | 31.82       | 36.92      | 40.66          |
|          | <b>RMM</b> ( $p = 2$ ) | 62%            | 54.11       | 79.19       | 90.60        | 74.49       | 87.46      | 77.17          |

Table 6: Per-task numbers behind Contribution 1 (PT-SVD): GLUE performance when merging five RoBERTa-base models compressed with PT-SVD at ranks  $r = 16, 32, 64, 128$ . As with LoRA (Table 5), the baselines degrade sharply while RMM ( $p = 2$ ) stays close to the individual models.